SUPPORTING ADULT GUIDANCE

# Expanding three brackets

Algebraic Expressions and Identities · Lesson 3. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

## What this lesson does

This lesson extends expanding to three brackets. Learners first see why there are 8 products by picking one term from each bracket, then use a two-stage grid: a 2 by 2 grid for the first pair, then a 3 by 2 grid with the third bracket. For all-positive brackets, an optional isometric box shows the expansion as a volume cut into 8 pieces, which can be pulled apart. A stretch proves the structure of $(x + a)(x + b)(x + c)$, and the Creativity panel uses it to predict coefficients and solve mystery brackets.

Driving Question: *What happens when three brackets multiply?*

## The mathematics, in plain terms

$(x + a)(x + b)(x + c) \equiv x^3 + (a + b + c)x^2 + (ab + bc + ca)x + abc$. The expansion is the sum of the $2^3 = 8$ products formed by choosing one term from each bracket. When a bracket has a leading coefficient other than 1, the leading coefficient of the cubic is the product of the leading coefficients, and the quick checks must be adjusted.

## The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

### Example 1 — Two stages

Expand and simplify $(x + 1)(x + 2)(x + 3)$.

**Stage 1: $(x + 1)(x + 2)$**

| × | $x$ | **2** |
|---|---|---|
| $x$ | $x^2$ | $2x$ |
| **1** | $x$ | 2 |

**Stage 2: $(x^2 + 3x + 2)(x + 3)$**

| × | $x$ | **3** |
|---|---|---|
| $x^2$ | $x^3$ | $3x^2$ |
| **3**$x$ | $3x^2$ | $9x$ |
| **2** | $2x$ | 6 |

| | |
|---|---|
| **1.** Stage 1: fill the 2 by 2 grid, then collect. | $x^2 + 2x + x + 2 = x^2 + 3x + 2$ |
| **2.** Stage 2: $x^2 + 3x + 2$ goes down the side; $(x + 3)$ gives the column headings. Fill every cell. | six cells |
| **3.** Collect: like terms sit on the diagonals (the shaded $x^2$ cells). | $x^3 + 6x^2 + 11x + 6$ |
| **4.** Check: $1 \times 2 \times 3 = 6$, and $1 + 2 + 3 = 6$. | ✓ |

**$(x + 1)(x + 2)(x + 3) = x^3 + 6x^2 + 11x + 6$.**

**Example 2 — With negative numbers**

Expand and simplify $(x - 2)(x + 3)(x + 1)$.

**Stage 1: $(x - 2)(x + 3)$**

| × | $x$ | **3** |
|---|---|---|
| $x$ | $x^2$ | $3x$ |
| **–2** | $-2x$ | –6 |

**Stage 2: $(x^2 + x - 6)(x + 1)$**

| × | $x$ | **1** |
|---|---|---|
| $x^2$ | $x^3$ | $x^2$ |
| $x$ | $x^2$ | $x$ |
| **–6** | $-6x$ | –6 |

| | |
|---|---|
| **1.** Stage 1: fill the grid, keeping each sign with its term, then collect. | $x^2 + 3x - 2x - 6 = x^2 + x - 6$ |
| **2.** Stage 2: $x^2 + x - 6$ down the side, $(x + 1)$ as the column headings. | six cells |
| **3.** Collect along the diagonals. | $x^3 + 2x^2 - 5x - 6$ |
| **4.** Check: $(-2) \times 3 \times 1 = -6$, and $-2 + 3 + 1 = 2$. | ✓ |

**$(x - 2)(x + 3)(x + 1) = x^3 + 2x^2 - 5x - 6$.**

## One method, used twice

No new rule is introduced: the grid from Lesson 2 is used twice. The 'pick one from each' activity explains the count of 8 before any expanding, the box gives a picture for positive cases, and the quick checks (number term and $x^2$ coefficient) let learners verify their own answers. The box is optional and static; it is unavailable when any term is negative.

## The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

| Cornerstone | Its role here | Invitational timing |
|---|---|---|
| Connection | Powers meet brackets: $x \times x^2$ and a two-bracket warm-up. | ~5 min |
| Movement | Pick one from each: find all 8 products. | ~7 min |
| Reflection | The two-stage grid, with the optional box. Stretch: prove the structure of the coefficients. | ~18 min |
| Creativity | The coefficient detective: predict, then solve mystery brackets. | ~10 min |
| Rest | 1, 3, 3, 1: counting the pieces of the box. | ~2 min |

| | | |
|---|---|---|
| Nutrition | Intellectual: scaling up a method you already trust. | ~3 min |

## Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

**Writing $x \times x^2 = x^2$ or $2x^2$.**
Ask: “How many $x$s are multiplied together?”

**Multiplying all three brackets at once and missing products.**
Ask: “How many products should there be? Have you got 8?”

**Sign slips with negative numbers in stage 2.**
Ask: “What is the sign of each cell before you collect?”

**Collecting unlike terms ($x^3$ with $x^2$).**
Ask: “Are those the same power of $x$?”

**$(x + 2)^3 = x^3 + 8$.**
Ask: “How many brackets is that? Draw the grids.”

## Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

**Please hold this line**
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

## Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: expanding two brackets with the grid, and multiplying negative numbers. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

## Supporting through questions: an example

Suppose a learner gets $x^3 + 6x^2 + 11x + 5$ for $(x + 1)(x + 2)(x + 3)$. The aim is a question that lets them find the slip themselves.

- “What should the number term be, without expanding?”
- “Which cell of your stage 2 grid gives the number term?”

- “What is 2 × 3?”
- “Can you check the $x^2$ coefficient the same way?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

## Questions you might have

**Does it matter which two brackets are expanded first?**
No: multiplication can be done in any order. Choosing the easiest pair first can help.

**Why is the box not available for every question?**
A length cannot be negative, so the box only works when every term is positive. The grid works for everything.

**Is the proof required?**
It is a stretch. The secure core is the two-stage grid and the number-term check.

**What comes next?**
Lesson 4: simplifying algebraic fractions — when cancelling is allowed.

## Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

## Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

| Where they are | What it can look like |
|---|---|
| Approaching Proficiency | Beginning to engage; names some features and makes a start with support. |
| Developing Proficiency | Carries out the method with prompts; can explain part of the reasoning. |
| Proficiency | Works through independently and explains why each step is true. |

| | |
|---|---|
| Refining Proficiency | Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else. |

## A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

## If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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